

Probabilistic-Origin Framework (POF)

Unified Collapse, Higgs Duality, Forces, and Simulations

Comprehensive Whitepaper

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Project: Neural Optical Co-Processor

Fundamental Theorem of POF

Fundamental Theorem of POF

The Probability Operator and the Higgs Operator are dual, like integration and differentiation.

****Probability Operator (Integral Form):****

$$P_{\{s,o\}} = |s\rangle\langle s| \otimes |o\rangle\langle o|$$

Collapse as marginalization:

$$P(s|o) = P(s,o) / \sum_{\{s'\}} P(s',o)$$

****Higgs Operator (Derivative Form):****

Collapse as alignment with the vacuum expectation origin (vev).

****Theorem:****

Higgs Collapse (Derivative) $\leftrightarrow$ Probability Collapse (Integral)

Implication: Collapse is both accumulation and localization. The Higgs boson is the p

Unification of the Four Forces

Four Forces in POF

- **Gravity**: Collapse of spacetime origin → consistent metric.
- **Electromagnetism**: Collapse of phase origin → U(1) gauge slice.
- **Weak Interaction**: Collapse of isospin origin → SU(2), selects $W^\pm$, Z.
- **Strong Interaction**: Collapse of color origin → SU(3), confinement.

Each force = relational rule applied to a different origin.

Unification: forces are emergent origin collapses.

Higgs Operator as Central Collapse Hub

Higgs as Collapse Hub

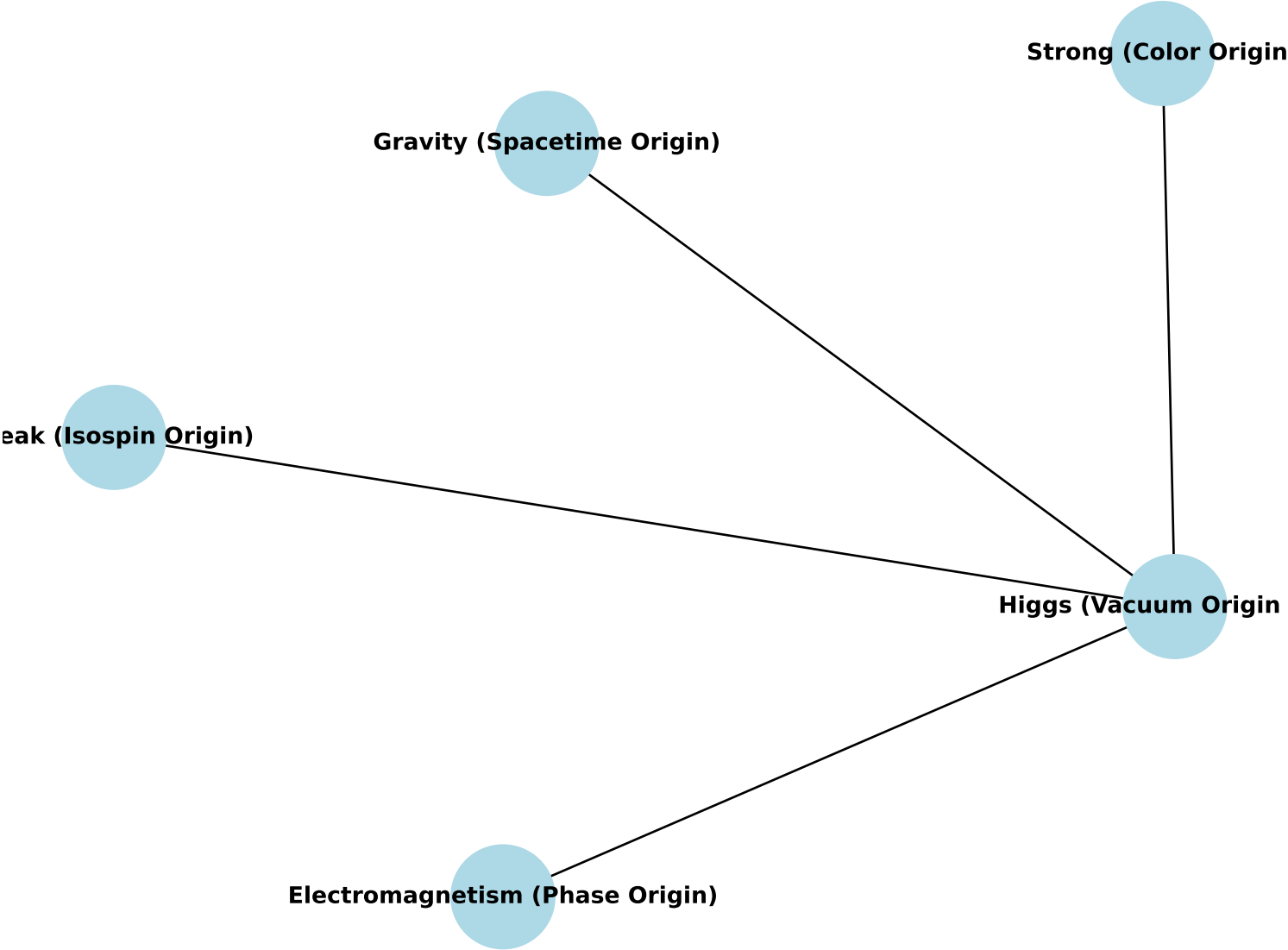
- Standard Model: Higgs gives mass via $\text{vev} \approx 246 \text{ GeV}$.
- In POF: Higgs = probabilistic origin in field space.
- Higgs boson = ripple of origin collapse.

Central role:

- Links gravity, EM, weak, strong collapses.
- Acts as anchor for the unified relational collapse rule.

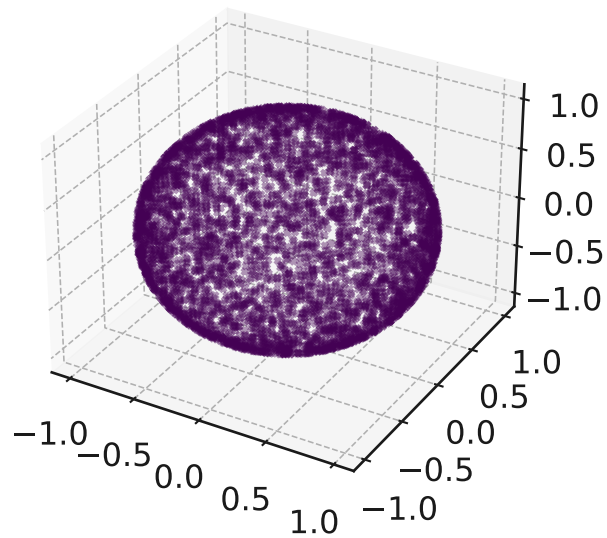
Diagram below illustrates the Higgs as the central hub where all force-origin collapses

Higgs as the Central Collapse Hub in POF

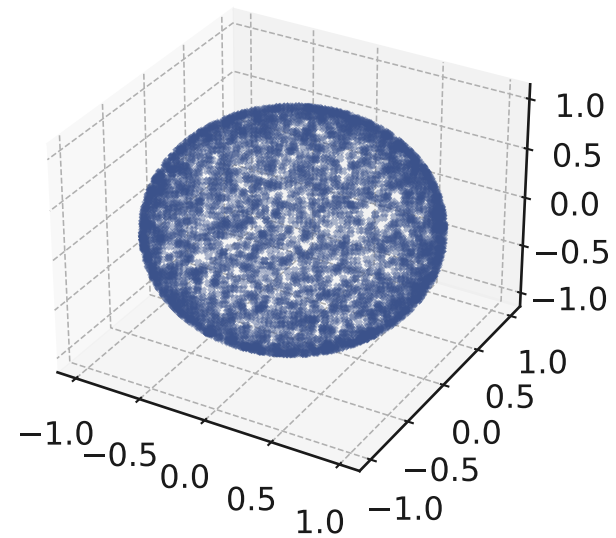


SU(2) Collapse Trajectories (Bloch Sphere)

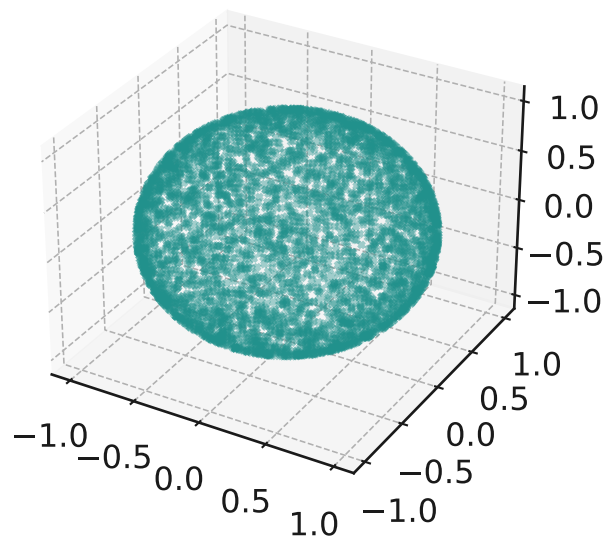
SU(2) Collapse t=0/20



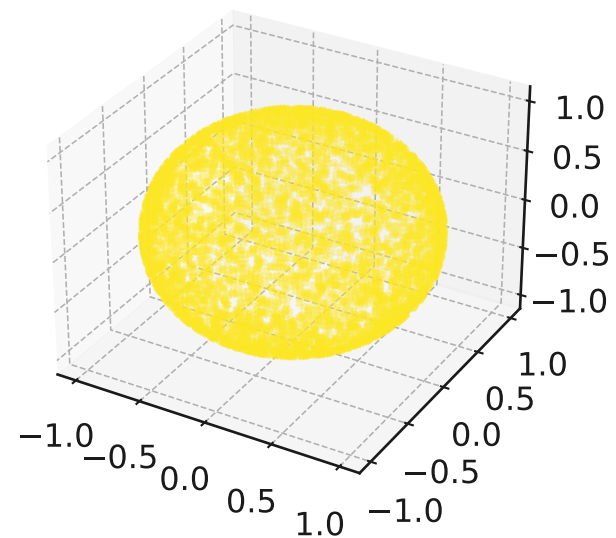
SU(2) Collapse t=5/20



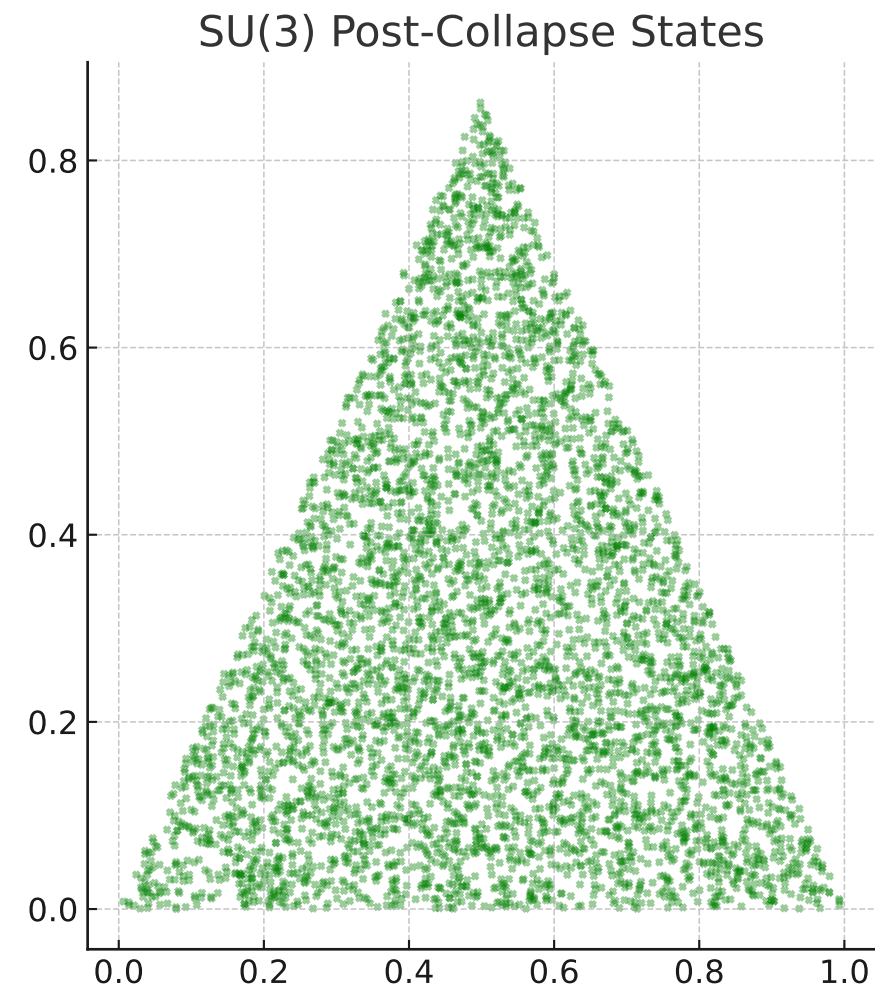
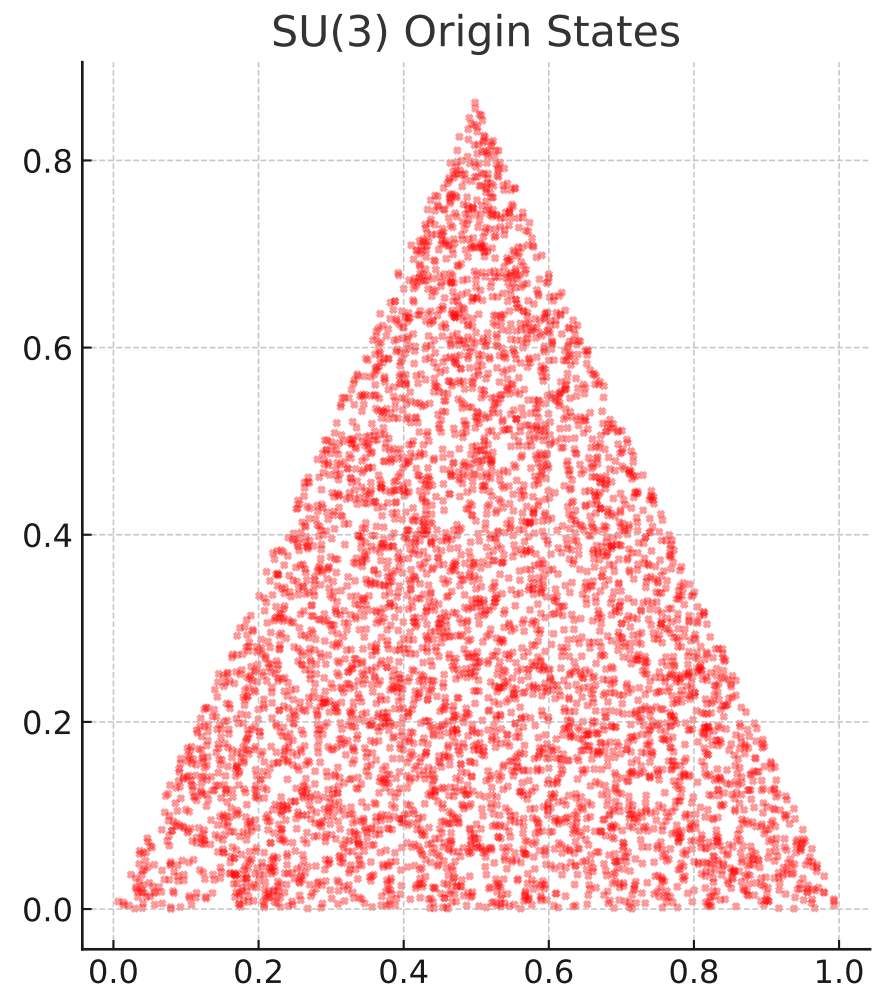
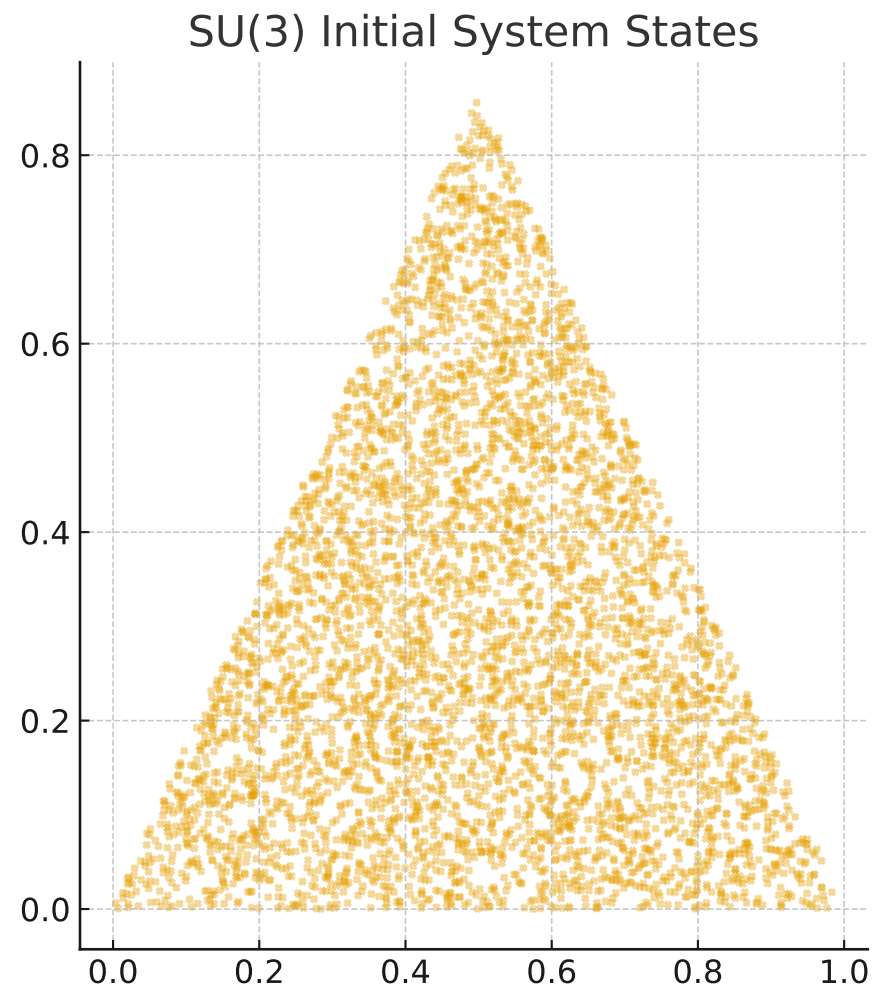
SU(2) Collapse t=10/20



SU(2) Collapse t=20/20

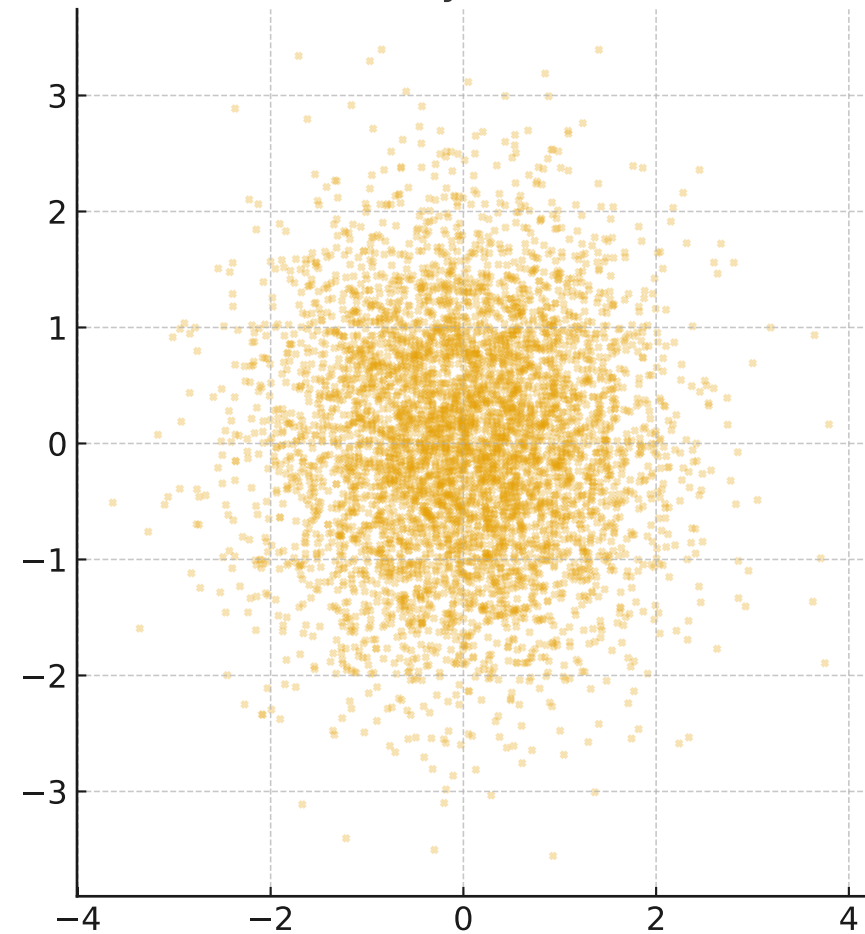


SU(3) Collapse in Probability Simplex

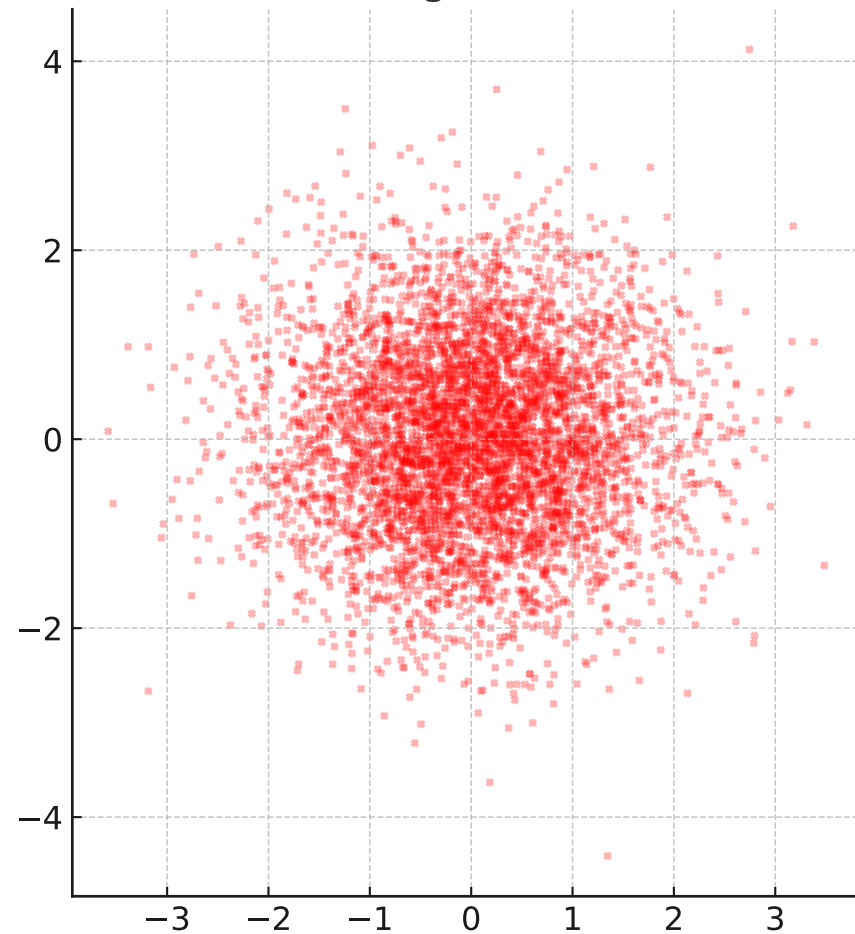


SO(3,1) Collapse in Relativistic Space

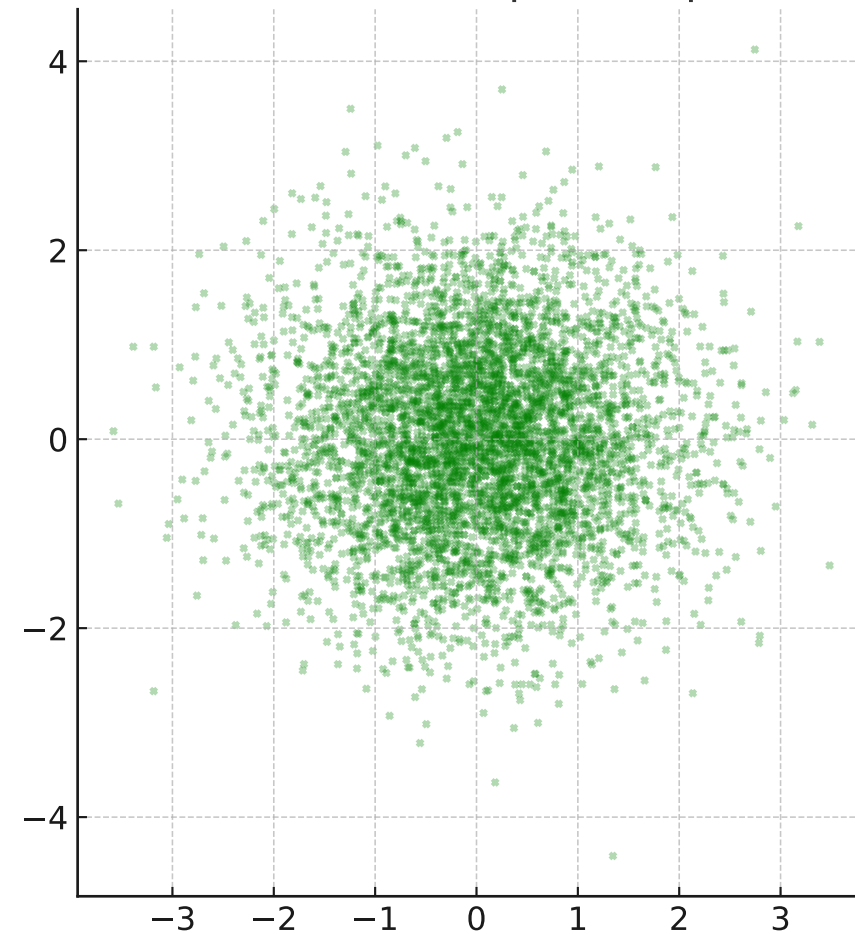
SO(3,1) Initial System Events (t-x)



SO(3,1) Origin Events (t-x)



SO(3,1) Post-Collapse (t-x plane)



Conclusion: POF as Unified Framework with Simulation Evidence

Conclusion

The Fundamental Theorem of POF unites the probability operator (integral form) and the

The Higgs boson is the hub where these collapses converge. Calibration in engineered

Implication: POF offers a unified foundation bridging quantum mechanics, the Standard